Mst Afsana Mimi

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Summary

Statistics master's graduate with experience cleaning and analyzing large datasets, building reproducible workflows, and communicating quantitative findings. Proficient in SQL, Excel, Python, R, Tableau, and Power BI, with strengths in data validation, trend analysis, statistical modeling, and reporting. Prepared to support business teams with accurate analysis and clear, actionable insights.

Education

University of Massachusetts Amherst
Master's in Statistics (CGPA: 3.77) Sep 2023 – May 2026

University of Dhaka
Master's in Applied Statistics (CGPA: 3.75) Jan 2017 – Dec 2018

University of Dhaka
Bachelor's in Applied Statistics (CGPA: 3.61) Jan 2013 – Dec 2016

Technical Skills

Analytics & Business Intelligence

- SQL (joins, aggregations, window functions), Excel, Python (Pandas, NumPy), R
- Tableau, Power BI, data visualization, dashboard development, and quantitative reporting

Data Analysis & Quality

- Data collection, cleaning, transformation, integration, and dataset construction
- Data validation, consistency checks, trend analysis, and attention to data integrity
- Regression, hypothesis testing, statistical modeling, and model evaluation

Workflows & Platforms

- ETL/ELT workflows, reproducible analysis, Git/GitHub
- AWS (S3, EC2) – familiarity

Experience

Teaching Assistant (Data & Statistics)
UMass Amherst Sep 2023 – May 2026

- Supported student projects involving R/Python-based data analysis.
- Explained statistical results clearly and helped students turn analysis into evidence-based conclusions.

Research Assistant (NLP & Data Analysis)
University of Dhaka Jul 2021 – Jun 2022

- Collected, cleaned, and transformed large-scale text data through reproducible ingestion and preprocessing workflows.
- Used Python to analyze patterns, engineer features, and evaluate machine-learning models.
- Performed validation and consistency checks to improve data accuracy and the reliability of reported findings.

Research Assistant (Socio-economic Data Analysis)
University of Dhaka Jul 2018 – Jun 2019

- Cleaned, combined, and structured large-scale survey datasets for reproducible analysis.
- Applied statistical methods to identify socio-economic trends and decision-relevant findings.
- Communicated analytical results for policy-focused research and maintained workflows with Git.

Projects

Can Agentic AI Eliminate the Need for a Dedicated Relational Foundation Model?

- Developing an analytical framework for relational databases that interprets schemas and constructs evidence across linked tables.
- Evaluating prediction workflows and designing a benchmark for realistic structured-data tasks.

Gaussian Process Modeling with Fairness Constraints

- Developing Gaussian Process models with fairness constraints across demographic groups.
- Designing evaluation strategies to assess trade-offs between predictive accuracy and bias.

High-Dimensional Feature Selection (MCAR)

- Developed a feature-selection method for high-dimensional censored data and evaluated it through simulations and biomedical data.
- Compared performance with established approaches; results were published in *Statistics in Medicine*.

Awards and Scholarship

- Conference Awards for Scientists (CASc) by International Society for Clinical Biostatistics (ISCB).
- National Science and Technology (NST) Fellowship Award by Education Board of Bangladesh.
- Dhaka University Merit Scholarship for academic results in B.S. Hon's examination.

Publications

- Mimi, A., & Khan, M. H. R. (2021). Variable selection for censored data using modified correlation adjusted correlation (mcar) scores. Statistics in Medicine, 40(23), 5046–5064.

Google Scholar: https://scholar.google.com/citations?user=LaVyjwYAAAAJ&hl=en&oi=sra